Curriculum Vitae

Personal Information

Name:	Gang Liu	Date of Birth:	Mar. 1993
Ethnicity:	Han Chinese	Gender:	Male
Native Place:	Longnan, Gansu	Marital Status:	Married
Phone:	15868721396	Email:	gangliuphys@gmail.com
Institution:	Lanzhou University	Degree:	Ph.D.



Education

Sep. 2020 – Jun. 2024	Lanzhou University	Theoretical Physics	Ph.D. in Physics
Sep. 2016 – Jun. 2019	Wenzhou University	Theoretical Physics	M.Sc. in Physics
Sep. 2012 – Jun. 2016	Longdong University	Physics Education	B.Sc. in Physics

Major Courses: General Physics, Four Fundamental Courses of Physics, Quantum Optics, Group Theory, Advanced Quantum Mechanics, Solid State Theory, Advanced Computational Physics, Advanced Statistical Physics, Laser Application Technology, etc.

Work Experience

Jul. 2024 – Jun. 2026	Zhejiang University	Postdoctoral Fellow
Jul. 2026 – Present	Zhejiang University	Postdoctoral Fellow

Research Interests

My research focuses on fundamental problems in quantum physics, including quantum optics, quantum precision measurement/quantum metrology, quantum phase transitions, and the preparation of macroscopic quantum states. Within the framework of light–matter interactions and hybrid quantum systems, I investigate the preparation and manipulation of nonclassical states, the control mechanisms of quantum phase transitions, and the applications of quantum resources in quantum precision measurement and quantum error correction. My research interests include:

1. Preparation of collective atomic spin-squeezed and entangled states in cavity QED systems;
2. Control mechanisms of quantum phase transitions in the quantum Rabi model and its extensions;
3. Critical quantum metrology based on quantum critical systems;
4. Preparation of nonclassical magnon states, including squeezed states, cat states, and Fock states, in magnon-based hybrid quantum systems;
5. Theoretical schemes for preparing magnonic Gottesman–Kitaev–Preskill (GKP) states and their applications in bosonic quantum error correction.

Computer Skills

Programming Languages: Mathematica, Python, MATLAB, etc.

Software: Microsoft Office, LaTeX, 3ds Max, Adobe Illustrator, etc.

Research Experience

Jan. 2027 – Dec. 2029, National Natural Science Foundation of China (NSFC) Youth Program, Realization and Applications of Superradiant Phase Transition in Magnon-Superconducting Qubit Systems, **Principal Investigator**

Jan. 2025 – Dec. 2027, Zhejiang Provincial Natural Science Foundation for Distinguished Young Scholars, Quantum State Preparation Based on Magnon Hybrid Quantum Systems, Participant

Jul. 2024 – May 2026, National Key Research and Development Program/NSFC Major Research Plan, Construction and Manipulation of Solid-State Hybrid Quantum Systems Based on Magnons, Participant.

Jan. 2020 – Dec. 2024, National Natural Science Foundation of China (NSFC) General Program, Geometric Control of Quantum Phase Transitions in Rashba Spin–Orbit Coupled Nanostructures, Participant

Jan. 2016 – Dec. 2019, National Natural Science Foundation of China (NSFC) Youth Program, Research on the Generation of Highly Spin-Squeezed States in Atomic Ensembles, Participant

Publications

(* *these authors contributed equally*; † *corresponding author*)

[1] Z.-X. Lu*, X. Zuo*, X.-L. Hei*, **G. Liu**†, Z.-X. Liu†, Q. Guo†, P.-B. Li†, J. Li†. *Quantum Magnonics: Quantum States Generation and Applications*, arXiv:2607.27771 (2026)(Invited Topical Review in *Quantum Sci. Technol.*).

[2] G. Li*, **G. Liu***,†, R.-C. Yang, J. Li†. *Preparing two-mode magnonic Schrödinger cat states in a cavity-magnon-qubit system*, arXiv:2606.25511 (2026) (submitted to *PRA*).

[3] R.-C. Yang*, **G. Liu***, G. Li, J. Li†. *Macroscopic entanglement between two magnon modes via two-tone driving of a superconducting qubit*, Phys. Rev. A 114, 043707 (2026)

[4] Z.-X. Lu*, **G. Liu***, M. Fadel, J. Li†, *Magnonic Gottesman-Kitaev-Preskill states*, arXiv:2604.27565 (2026) (submitted to *PRL*).

[5] **G. Liu**†, J. Liu, F. Qiao†, R.-C. Yang†. *Generation of magnonic squeezed state and its superposition in a hybrid qubit-magnon system*, arXiv:2604.02924 (2026) (submitted to *PRA*).

[6] **G. Liu**, G. Li, R.-C. Yang, W. Xiong†, J. Li†. *Magnon squeezing near a quantum critical point in a cavity-magnon-qubit system*, Phys. Rev. A 113, 033707 (2026).

[7] **G. Liu**, W. Xiong†. *A^2-robust superradiant phase transition in hybrid qubit-cavity optomechanics*, Phys. Rev. A 112, 063727 (2025).

[8] **G. Liu**†, G. Li, H. Tang†, J. Li. *Magnon cat states in a cavity-magnon-qubit system via two-magnon driving and dissipation*, Phys. Rev. A 112, 023709 (2025).

[9] **G. Liu**, W. Xiong†, Z.-J. Ying†. *Switchable superradiant phase transition with Kerr magnons*, Phys. Rev. A 108, 033704 (2023).

[10] **G. Liu**, Y.-N. Wang, L.-F. Yan, N.-Q. Jiang, W. Xiong, M.-F. Wang†. *Spin squeezing via one- and two-axis twisting induced by a single off-resonance stimulated Raman scattering in a cavity*. Phys. Rev. A 99, 043840 (2019).

Collaborative Publications

[11] J. Liu, Y. Mu, L. Song, **G. Liu**, M. Wang. *Engineering near-unitary one-axis-twisting evolution via a driven Tavis-Cummings model*. Phys. Rev. A 113, 043710 (2026).

[12] L. Wang, Y. Wang, Y. Cheng, Z. Yan, L. Xie, **G. Liu**, J. Fan, D. Wang, Y. Song, L. He, W. Xiong, M. Wang. *Entangling spins using cubic nonlinear dynamics*. arXiv:2301.04520 (2023).

[13] Z.-J. Ying, S. Felicetti, **G. Liu**, D. Braak. *Critical Quantum Metrology in the Non-Linear Quantum Rabi Model*, Entropy 24(8), 1015 (2022).